

Known: $E = (8.5 - 11.5) \times 10^6 \text{ psi}$, $300 \text{ lb}_f \leq F \leq 500 \text{ lb}_f$,
max $\delta = 0.0009 \text{ in}$, $\delta = \frac{FL}{AE} \rightarrow L = \frac{\delta AE}{F}$, $A = \frac{\pi d^2}{4}$
Aluminum Bar

Unknown/chosen: $F = 300 \text{ lb}$, $d = 1 \text{ in}$, $E = 11.45 \times 10^6 \text{ psi}$

$$A = \frac{\pi (1 \text{ in})^2}{4} = 0.79 \text{ in}^2, \quad L = \frac{0.009 \text{ in} (0.79 \text{ in}^2) (11.45 \times 10^6)}{300 \text{ lb}}$$

$$\sigma = \frac{F}{A} = \frac{300 \text{ lb}}{0.79 \text{ in}^2} = 381.97 \text{ psi} \quad \frac{L = 269.7849 \text{ in}}{\text{or } 0.38197 \text{ ksi}}$$

$$\text{SF} = \frac{40 \text{ ksi}}{0.38197 \text{ ksi}} = 104.72, \quad \text{Weight} = 0.79 \text{ in}^2 (269.7849 \text{ in}) (0.098 \text{ lb/in}^3) \\ = 20.765 \text{ lb}$$

Design Reflection

$$\delta = \frac{300 \text{ lb} (269.7849 \text{ in})}{(0.79 \text{ in}^2) (11.45 \times 10^6 \text{ psi})} = 0.009 \text{ in (given)} \checkmark$$

δ from FEA is 0.00905 in

$$\text{Discrepancy} = \frac{0.00905 - 0.009}{0.009} \times 100 = \frac{0.00005}{0.009} \times 100 = 0.56\% \\ 905.56\%$$